

Spectral: Expander Graphs for Service Discovery

Service discovery is a core component of distributed systems and microservice architecture, automating the process by which microservices locate and communicate with one another to manage health checks and trigger service requests. The predominant approach uses a centralized service registrar (a dedicated registry), which acts as a lookup directory for services. However, this centralized model introduces several critical downsides: Single Points of Failure (SPOFs), scaling bottlenecks, and high sensitivity to churn, where services frequently register or deregister.

We propose **Spectral**, a peer-to-peer service discovery system inspired by expander graph topology. Spectral distributes membership information across nodes in a sparse but highly connected network, reducing single points of failure and eliminating centralized coordination bottlenecks from traditional approaches.

An expander graph is a sparse graph with strong connectivity properties. It is characterized by rapid information diffusion and resiliency to random node failures. In Spectral's context, the expander structure enables efficient propagation of service registration, deregistration, and metadata under high churn, while maintaining bounded node degree and low communication overhead.

Rather than relying on a fixed global topology, Spectral treats expansion as a dynamic property that changes as nodes churn. Nodes continuously rewrite their connections through peer sampling and membership gossip protocols. Its goal is to constantly maintain ideal connectivity and fast mixing properties over time. This makes Spectral resilient to service/node churn and relieves the scalability limitations of current service registries.

Local Expander Graphs Construction

<https://people.irisa.fr/George.Giakakopoulos/papers/stoc22flip.pdf> (Expansion on paper below)

<https://arxiv.org/pdf/1510.07768> (Uses flip chains to converge to expander)

<https://ieeexplore.ieee.org/document/1209234>

<https://arxiv.org/pdf/1206.1522> (Self healing expanders—for churn)

Similar Applications/Systems

<https://github.com/hashicorp/hyprview>

<https://developer.hashicorp.com/consul>

Spectral Gap Estimation Technique(s)

<https://hal.science/hal-03942606v1>

<https://arxiv.org/html/2511.03291> (But we'd modify this for distributed systems)